

IHAB HACHEM

Senior Backend & Software Engineer | Java & Spring Boot | PHP & Laravel | Distributed Systems & AI Engineering

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PROFESSIONAL SUMMARY

Backend engineer with experience building and modernizing REST APIs and backend architecture across Java/Spring Boot and PHP/Laravel. Current work centers on migrating legacy Laravel services to Spring Boot, designing relational database schemas, and using views and stored procedures to keep heavy operations out of the application layer. Background also includes Python/FastAPI microservices, Docker-based deployment on Linux, and system documentation for complex integrations. Alongside core backend work, has designed and built a self-hosted Retrieval-Augmented Generation (RAG) assistant and contributed to AI assistant architecture involving MCP-based tooling — applying AI as practical backend infrastructure rather than as a separate research track. Comfortable taking ownership of an existing codebase, understanding its structure, and evolving it toward a cleaner, more maintainable design.

PROFESSIONAL EXPERIENCE

Backend Engineer — **Moobitek**, Beirut, Lebanon

April 2024 – Present

Working within an existing PHP Laravel codebase while leading the migration of core services to Java Spring Boot, building new API versions with cleaner structure and optimized query patterns.

- Design and build new REST APIs in both Laravel and Spring Boot, and maintain existing endpoints as requirements evolve.
- Contribute to the migration of legacy Laravel modules to Spring Boot, restructuring business logic into a maintainable service layer.
- Analyze and document the legacy system's structure, including database schema and API flow diagrams for complex processes.
- Design relational database schemas and use MyBatis alongside views and stored procedures to move heavy operations out of the application layer.
- Build and manage Docker containers for services, and handle deployment and testing on Linux servers.
- Write API documentation to support handoff between backend, frontend, and integration teams.
- Built and deployed an internal MCP Gateway in Spring Boot 4 / Java 21 — a modular service hosting multiple independent MCP modules under shared infrastructure, with per-developer API keys and systemd-managed deployment.
- Lead the development of an annuity application using Spring Boot 4 and Java 21, driving backend architecture and API design while integrating MySQL for persistence, Redis for caching, and Docker/Nginx for containerized deployment.

Python Backend Developer — **Auto4Utos**, United Kingdom (Remote)

November 2022 – August 2023

Built and maintained REST APIs for a new application using FastAPI, with responsibility for testing, deployment, and supporting infrastructure.

- Built and maintained REST APIs using FastAPI, validating each endpoint with Postman before release.
- Wrote automated unit tests to cover API behavior and reduce regressions during ongoing development.
- Integrated Docker microservices into the application and deployed them to a Linux server for the wider team.
- Set up Elasticsearch, Kibana, and Logstash for centralized application logging and monitoring.
- Used a Machine Learning Kit for automated document validation, and worked within an Agile/Jira workflow.

AI ENGINEERING & INTELLIGENT BACKEND SYSTEMS

Practical, engineering-focused AI work: building AI-powered backend services, RAG pipelines, and AI assistants rather than research.

- Designed and built a self-hosted RAG AI assistant (open-source, GitHub) using Java 17, Springboot 3, and LangChain, with Hugging Face Transformers for embeddings and VectorDB for vector search over document-based knowledge, code base and databases through MCP.
- Backed the assistant with MySQL for structured data, containerized the full stack with Docker and Docker Compose, and served it behind Nginx via REST APIs — built to be integrated into business applications.
- Contributed to an AI assistant architecture combining RAG, database access, Jira integration, and MCP-based tools, connecting LLM-driven workflows to existing backend systems.
- Built a modular MCP Gateway in Spring Boot to host multiple independent MCP modules under shared infrastructure, enabling AI tool integrations (including Claude Desktop connectivity) across internal services.

SELECTED PROJECTS

Selected projects showcasing backend engineering, system architecture, AI integration, and the development of scalable software solutions across diverse domains.

RAG AI Assistant:

Java, Spring Boot 3, JDK 17, Vector Database

Designed and developed a self-hosted Retrieval-Augmented Generation (RAG) AI assistant using Java and Spring Boot 3, enabling document-based knowledge retrieval and AI-powered question answering. Integrated vector database capabilities for semantic search and designed the application as a backend service that can be integrated with other business applications.

MCP Gateway & AI Tooling:

Java, Spring Boot, MCP

Designed and developed a modular MCP Gateway to host multiple independent MCP modules under shared infrastructure, enabling AI assistants to interact with backend services and external tools through a unified architecture. The design supports adding new MCP modules without requiring changes to the core infrastructure.

Multi-Service & E-commerce Platforms:

PHP, Laravel, REST APIs

Developed backend services for multi-service and e-commerce applications, including MSM, Multivendor E-commerce, and E-commerce Backend platforms. Designed REST APIs for mobile and administrative clients covering users, products, inventory, orders, notifications, permissions, and location-based services. Integrated authentication, payment gateways, SMS and email services, and configured server-side deployment with SSL and production subdomains.

Multivendor E-commerce Admin Panel:

React, TypeScript

Developed a React/TypeScript business administration dashboard for managing categories, companies, products, users, notifications, and dynamic homepage banners.

Restaurant Delivery Application:

React Native

Developed the client and driver applications for a restaurant delivery platform, including product browsing, cart and order management, driver assignment, location tracking, delivery status updates, and profile management.

Earlier Projects:

Developed additional applications including a Digital Menu platform, Car Management System, Tulip POS, and Library Management System, covering backend APIs, relational databases, inventory management, order processing, administration, and client-facing applications using PHP, C#, Java, and React Native.

TEACHING & MENTORING

- Full Stack Instructor, CIS College (2022) & AUB (2021) — Taught Java, Python, OOP, data structures, REST API design, and PHP/C# backend development to junior developers.
- Private Advanced Java Instructor (2018) — Taught JavaFX UI development, database integration, and file handling.

TECHNICAL SKILLS

Programming languages: Java, Python, PHP, C#, JavaScript, TypeScript.

Backend & Frameworks: Spring Boot, Laravel, FastAPI, Spring Security, REST APIs, Microservices.

System Architecture: Distributed Systems, System Design, Backend Architecture, API Design, API Documentation.

AI Engineering: RAG, AI Assistants, LLM Integration, MCP, Vector Databases, Vector Search, LangChain, Hugging Face Transformers.

Databases & Persistence: MySQL, MariaDB, Microsoft SQL Server, SQLite, MyBatis, Relational Database Design, Views, Stored Procedures.

Caching & Messaging: In-memory caching, Redis, Solr, RabbitMQ, Kafka.

Integration & Communication: Twilio, Payment Gateway Integration, SMS & Email Services, External API Integration.

Business Rules: Drools.

Security & Authentication: Spring Security, JWT, Laravel Sanctum, API Authentication, API Keys.

DevOps & Infrastructure: Docker, Docker Compose, Nginx, Linux, Git, GitHub, server deployment.

Observability & Logging: Elasticsearch, Kibana, Logstash.

Testing & API Tools: Postman, Unit Testing, Automated API Testing.

Frontend : React, React Native, Vue.js, HTML5, CSS3

Tools & Workflow: Jira, Agile

EDUCATION

M.S. in Computer and Communication Engineering — **AUL — Arts, Sciences and Technology University**, Beirut, Lebanon

October 2021 – February 2023

Thesis on a digital identity platform designed to replace physical ID documents.

LANGUAGES

Arabic — Native • English — Fluent • French — Fluent